FPGA Parallelism - Lab 1

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HLS synthesized latency:

```
================================================================
== Performance Estimates
================================================================
+ Timing:
    * Summary:
    +--------+---------+----------+------------+
    |  Clock |  Target | Estimated| Uncertainty|
    +--------+---------+----------+------------+
    |ap_clk  | 10.00 ns| 7.300 ns|     2.70 ns|
    +--------+---------+----------+------------+

+ Latency:
    * Summary:
    +---------+---------+----------+----------+-------+-------+---------+
    |  Latency (cycles) | Latency (absolute) |    Interval   | Pipeline|
    |   min   |   max   |    min   |    max   |  min  |  max  |   Type  |
    +---------+---------+----------+----------+-------+-------+---------+
    |    52508|    52508| 0.525 ms| 0.525 ms|  52509|  52509|       no|
    +---------+---------+----------+----------+-------+-------+---------+
```

Co-simulation latency and clock period:

RTL	Status	Latency(Clock Cycles) min	Latency(Clock Cycles) avg	Latency(Clock Cycles) max	Interval(Clock Cycles) min	Interval(Clock Cycles) avg	Interval(Clock Cycles) max	Total Execution Time (Clock Cycles)
VHDL	NA	NA	NA	NA	NA	NA	NA	NA
Verilog	Pass	51499	51499	51499	NA	NA	NA	51499

Post-implementation resource utilization:

```
#=== Post-Implementation Resource usage ===
SLICE:             0
LUT:           27498
FF:            32641
DSP:               8
BRAM:            153
URAM:              0
LATCH:             0
SRL:             645
CLB:            5894

#=== Final timing ===
CP required:                    10.000
CP achieved post-synthesis:     3.994
CP achieved post-implementation: 6.269
Timing met
```

Final speedup ratio calculation:

51499 cycles * 6.269 post-imp. clock period = 0.322847 ms

6.8817251 / 0.322847 = <u>21.3157 speedup</u>